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Turning Flared Gas into Energy via Flared Gas Recovery System (FGRS)

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Abstract

Gas flaring remains a significant environmental and economic challenge in the oil and gas industry. Billions of cubic feet of gas are flared annually, resulting in greenhouse gas emissions, energy loss, and environmental negative impact. In response to global efforts to reduce emissions and enhance energy efficiency, Flare Gas Recovery Systems (FGRS) have emerged as a viable solution for capturing and reusing flare gases. Among various compression technologies, the Liquid Ring Compressor (LRC) stands out for its ability to handle wet, not processed, and variable-composition gases typical of flaring streams. This paper presents a comprehensive overview of flare gas recovery using liquid ring compressor. It begins with an examination of the sources and impacts of flare gas emissions, followed by an in-depth discussion of the LRC's operating principles, system design, and configuration. Additionally, the paper addresses operational considerations such as seal liquid management, and system reliability. By integrating technical design with regulatory context, this paper highlights the LRC-based FGRS as a sustainable and robust solution to reduce routine flaring in oil and gas operations.

Introduction

Gas flaring, the controlled burning of excess hydrocarbons, has been generally a routine practice in operations. Originally implemented as a safety measure to relieve pressure and prevent unintentional gas release, flaring has evolved into a routine operation due to limitations in gas infrastructure, market access, and storage capabilities (Elvidge et al., 2016). While flaring can be necessary under specific operational conditions, its continuous and large-scale occurrence presents critical challenges in terms of environmental sustainability and energy conservation.

According to the World Bank's Global Gas Flaring Reduction Partnership (GGFR), over 139 billion Cubic Meters (BCM) of gas were flared globally in 2022, releasing approximately 400 million tonnes of CO₂-equivalent emissions into the atmosphere (World Bank, 2023). This not only contributes to climate change but also results in the loss of valuable natural resources that could otherwise be captured and monetized. Moreover, flaring is often associated with the release of carbon, volatile organic compounds (VOCs), and other pollutants harmful to human health and ecosystems (Anejionu et al., 2015).

In response to increasing environmental awareness and stricter emission standards, oil and gas producers are under pressure to minimize routine flaring and improve resource utilization. One of the most promising

strategies to achieve this is through Flare Gas Recovery Systems (FGRS), which are designed to capture and compress flare gas for fuel or further processing. These systems not only reduce emissions but also create economic value by converting waste gas into a usable and energy source.

Among various FGRS technologies, the Liquid Ring Compressor (LRC) has gained particular attention for its adaptability to harsh operating conditions. LRCs are positive displacement compressors capable of handling gas streams with high moisture content, solids particles, and varying chemical compositions — all common characteristics of flare gas. Their simple design, low maintenance requirements, and capacity for oil-free compression make them especially suitable for continuous recovery operations (Twardowski & Iwaszenko, 2014).

This paper provides a detailed academic examination of flare gas recovery utilizing liquid ring compressor technology. The following sections describe the nature of flare gas and its environmental impacts, the general configuration of FGR systems, the technical characteristics of LRCs, system integration challenges, and the environmental benefits in the context of global regulatory frameworks. The objective is to offer a comprehensive understanding of how LRC-based FGRS can contribute to a more sustainable and efficient hydrocarbon industry.

Flare Gas and Environmental Challenges

Flare gas refers to the excess hydrocarbon gases that are combusted at flare stacks during oil and gas production, refining, or processing. It typically consists of methane, ethane, propane, hydrogen sulfide (H_2S), carbon dioxide (CO_2), and various heavier hydrocarbons. The composition varies depending on the source, gas processing means, and operational conditions (Rahimpour et al., 2012).

Flaring can be categorized as emergency, process, or routine. Emergency flaring occurs during upset conditions to prevent overpressure and ensure plant safety. Process flaring may occur during plant startup, shutdown, or maintenance. Routine flaring, on the other hand, results from capturing and utilizing the gas in different systems, such as purging and sealing. This last category is the most environmentally and economically problematic.

The environmental consequences of gas flaring are extensive. Firstly, combustion of hydrocarbons emits significant volumes of greenhouse gases (GHGs), primarily CO_2 and methane. Although methane has a shorter atmospheric lifetime, its global warming potential over 20 years is more than 80 times that of CO_2 (IPCC, 2021). Inefficient combustion in flaring can also result in methane slip, where unburned methane escapes into the atmosphere, exacerbating climate impact.

In addition to GHGs, flaring emits black carbon, a particulate byproduct of incomplete combustion. Black carbon has been identified as a short-lived climate pollutant that accelerates glacier melt and disrupts regional weather patterns (Bond et al., 2013). Moreover, flaring releases nitrogen oxides (NO_x), sulfur dioxide (SO_2), carbon monoxide (CO), and other pollutants that contribute to acid rain, respiratory illness, and ecosystem degradation (Stevens et al., 2017).

On a broader scale, continuous flaring contributes to visual pollution and socioeconomic discontent in communities near oil and gas facilities. Some regions have experienced health, agricultural, and social challenges linked directly to prolonged exposure to flare gas pollutants (Anejionu et al., 2015).

Reducing gas flaring, therefore, is not only an environmental priority but also a strategic opportunity for sustainable development in hydrocarbon-rich regions. This provides the context for adopting technologies like Flare Gas Recovery Systems (FGRS) that minimize emissions and enhance operational efficiency.

Flare Gas Recovery System Overview

Flare Gas Recovery Systems (FGRS) are engineered setups designed to capture, compress, and repurpose gas that would otherwise be flared. These systems offer a pathway to comply with regulatory emission limits, reduce environmental impact, and convert waste into economic value (Al-Mashaqbeh et al., 2019).

The typical FGRS consists of the following core components:

1. **Knockout Drum or Separator:** This vessel is installed upstream the compressor to remove condensates, slugs, and liquids from the flare gas. Effective separation is critical to prevent damage or inefficiency in the compressor.
2. **Compressor (e.g., LRC):** The heart of the recovery system, it pressurizes the low-pressure flare gas for storage or further processing. The choice of compressor is driven by gas composition, pressure profile, and required throughput.
3. **Seal Liquid System (in case of LRC):** Maintains the liquid ring necessary for compression. It may include a storage tank, pump, cooler.
4. **Gas System:** Once compressed, the recovered gas may be directed to gas header, fuel system, or gas processing units, such as glycol dehydration or amine sweetening.
5. **Control and Safety Systems:** Pressure control valves, emergency shutdown systems (ESD), and programmable logic controllers (PLC) ensure safe and stable operation.

FGRS is operated in continuous operation, capitalizing on flare gas volume and consistency. In upstream applications such as gas-oil separation plants (GOSPs) and gas plants, flare flow is often continuous and requires rapid system response.

The integration of FGRS into existing facilities depends on factors such as site layout, gas composition, backpressure constraints, and existing infrastructure. Despite these complexities, technological advancements, and modular solutions have improved the feasibility of deploying FGRS across varied oil and gas operations ([Alimoradi et al., 2014](#)).

In this context, the liquid ring compressor is emerging as a highly viable solution for compressing flare gas in recovery systems, especially where gas quality is variable and contamination is expected.

Liquid Ring Compressors (LRCs)

Liquid Ring Compressors (LRCs) are positive displacement machines that compress gas using a rotating impeller inside a casing partially filled with liquid, typically water. Unlike conventional compressors, the LRC uses a liquid to form a seal and create compression chambers, making it especially suitable for applications involving wet, not processed, or corrosive gases — such as those commonly found in flare systems ([Bloch & Hoefner, 1996](#)).

Operating Principle

An LRC consists of an eccentrically mounted impeller rotating inside a casing filled with a sealing liquid. As the impeller rotates, centrifugal force causes the liquid to form a rotating ring inside the casing. This creates a series of gas-filled pockets between the impeller blades and the liquid ring.

During operation, the eccentricity between the impeller and casing causes the volume of these pockets to vary. On the suction side, the pocket volume increases, drawing in flare gas. As the impeller continues to rotate, the pocket volume decreases, compressing the gas. The compressed gas is then discharged through the outlet port, while the seal liquid is recirculated ([Meher-Homji & Boyce, 2001](#)).

This method of compression allows the LRC to operate with near-isothermal characteristics, which is advantageous when compressing high-moisture or temperature-sensitive gases.

Advantages of LRCs in Flare Gas Recovery

LRCs offer several operational and technical advantages that make them ideal for flare gas recovery applications:

- **Liquid Handling Capability:** LRCs can tolerate entrained liquids or slugs in the incoming gas stream, reducing the risk of mechanical failure or efficiency loss. This is critical in flare gas systems, where gas may carry water vapor, hydrocarbons, and process condensates.
- **Contaminant Resistance:** Due to their simple internal construction and use of liquid seals, LRCs are less prone to wear from solid particles or corrosive gases compared to dry compressors like reciprocating or screw compressors.
- **Stable Compression for Variable Flows:** Flare gas volumes can vary significantly over time. LRCs can handle such fluctuations without the need for complex flow control systems or surge protection.
- **Durability and Simplicity:** LRCs have fewer moving parts, which results in lower maintenance demands, high reliability, and long service life under continuous duty.

Working Fluids and Materials

The choice of sealing fluid is crucial in LRC operation. In many cases, water is used due to its availability, low cost, and good thermal properties. However, for flare gas streams containing reactive gases (e.g., H_2S or CO_2), process-compatible oils or specially treated liquids may be used to prevent corrosion or chemical degradation. Material selection for LRC internals — such as stainless steel or coated carbon steel — depends on the corrosivity of the gas and the operating temperature. Elastomers used for sealing must also be compatible with hydrocarbons or acidic gases.

System Design and Configuration

The design of a Flare Gas Recovery System (FGRS) based on an LRC must consider several technical and safety factors to ensure effective, smooth, safe, and continuous operation. Key elements of such a system include pre-treatment units or separation drums, the compression unit, seal liquid systems, and downstream connections.

System Layout

A typical FGRS using a Liquid Ring Compressor consists of the following units arranged in sequence:

1. **Knockout Drum:** This vessel removes bulk liquids and slugs from the flare gas. It protects the LRC from hydraulic damage and reduces seal liquid contamination.
2. **Suction Scrubber (optional):** Provides further filtration of fine mists or particulates before the gas enters the compressor.
3. **Liquid Ring Compressor Package:** Includes the LRC itself, drive motor, and gearbox (if required). Often, it comes pre-skidded for ease of installation.
4. **Seal Liquid System:** Includes a reservoir (seal tank), pumps for liquid circulation, and heat exchangers for cooling the seal liquid. Filtration systems may also be added.
5. **Gas Discharge Line:** Directs the compressed flare gas to the gas header, low-pressure separator, or storage system.
6. **Control and Monitoring System:** Includes pressure and temperature transmitters, level sensors, vibration monitoring system, PLC controllers, and emergency shutdown systems (ESD). These ensure safe and smooth operation during variable flow and power fluctuations.

Design Considerations

1. **Pressure and Flow Matching:** The compressor must be sized to handle maximum expected flare flow while maintaining discharge pressure suitable for downstream use.
2. **Seal Liquid System:** Seal liquid must be monitored for temperature, contamination, and level. Heat buildup must be dissipated using coolers, such as air cooler or heat exchanger to avoid performance degradation.

3. **Backpressure Control:** Automatic valves and pressure regulators are used to maintain system stability and avoid compressor overload or surge during flare surges.
4. **Fire and Explosion Safety:** Since the gas handled is flammable, the system must comply with applicable fireproofing, area classification, and explosion-proof standards (e.g., ATEX, NFPA).
5. **Startup and Shutdown Procedures:** Prepared to prevent liquid hammer or pressure surges, especially in intermittent service. Pre-filling of seal liquid operation are often automated.

This configuration allows the system to function reliably even under wide variation in gas flow, which is characteristic of flare streams in upstream facilities and plants.

Operational Considerations

Successful implementation of a Flare Gas Recovery System (FGRS) using a Liquid Ring Compressor (LRC) depends heavily on operational reliability, safety, and adaptability to fluctuating process conditions. The inherent variability of flare gas volume and composition requires robust design and well-planned operational strategies.

Startup and Shutdown Procedures

Proper startup is essential to prevent mechanical damage or seal liquid displacement. The LRC system is typically started with the seal liquid system primed and operational. Suction and discharge isolation valves are checked for alignment, and motor-driven components are brought online in sequence. A soft start or variable frequency drive (VFD) is commonly used to reduce inrush current and mechanical stress.

During shutdown, it is important to flush the seal liquid system to remove contaminants. The compressor should be cooled gradually to avoid thermal shock. If the unit is expected to be out of service for an extended period, it should be drained and dried to prevent corrosion.

Seal Liquid Quality and Maintenance

The performance of an LRC heavily depends on the condition of the seal liquid. Over time, flare gas contaminants such as sulfur compounds, hydrocarbons, and particulates can accumulate in the seal liquid, leading to reduced efficiency and corrosion.

To address this, workers should implement:

1. Regular sampling and analysis of seal liquid.
2. Filtration systems to remove suspended solids.
3. Heat exchangers or air cooler to maintain optimal liquid temperature.
4. Automatic make-up systems to compensate for seal liquid losses.

Seal liquid replacement intervals should be based on both manufacturer recommendations and operational history.

Handling Slug Flow and Wet Gas

Although LRCs are well suited to handle wet gases and occasional slugs, excessive liquid loading can still impact performance. Knockout drums and scrubbers upstream of the compressor should be sized appropriately and maintained regularly to prevent liquid carryover.

Level instrumentation and alarms in the knockout drum are crucial. Sudden increases in liquid level may indicate upstream upset conditions and require rapid intervention to prevent flooding of the LRC.

Monitoring and Reliability

The following parameters should be continuously monitored during operation:

1. Suction and discharge pressure

2. Seal liquid temperature and level
3. Compressor vibration and bearing temperature
4. Motor current and power consumption

Modern systems integrate these sensors into a control system with programmable logic controllers (PLCs), enabling real-time diagnostics, remote monitoring, and automated shutdown in case of abnormalities.

Reliability-centered maintenance (RCM) strategies can be applied to ensure optimal uptime, particularly in critical operations where flaring must be avoided. Scheduled inspections of impeller clearance, seal condition, and wear parts are recommended.

Environmental and regulatory Context

The use of Flare Gas Recovery Systems has gained prominence due to increased regulatory scrutiny and global climate commitments. Several international frameworks and national policies now require or strongly encourage reduction of routine flaring.

World Bank Zero Routine Flaring Initiative

Launched in 2015, this initiative calls on governments and oil& gas companies to eliminate routine flaring by 2030. Signatory countries and corporations must develop plans to capture or utilize routine gas rather than flare it ([World Bank, 2023](#)). FGR systems using LRCs can be a key enabler of this target.

Environmental Compliance

Many countries now include flaring limits in environmental permits. Exceeding these limits can result in fines, operational restrictions, or legal action. Deploying LRC-based FGRS helps operators remain within environmental compliance while improving their sustainability profile.

FGR systems also support the attainment of emission reduction targets under frameworks like:

- Paris Agreement commitments
- GHG Protocol and Scope 1 emission control
- Corporate ESG (Environmental, Social, and Governance) metrics

Health and Safety Benefits

By capturing flare gas, FGRS also reduce ground-level emissions of harmful pollutants, including volatile organic compounds (VOCs), sulfur compounds, and particulate matter. This leads to improved air quality around facilities, enhancing worker safety and community relations.

Conclusion

The deployment of Flare Gas Recovery Systems using Liquid Ring Compressor technology represents a strategic solution to one of the oil and gas industry's most pressing environmental challenges. By capturing and repurposing flare gas, these systems reduce greenhouse gas emissions, conserve energy, and contribute to regulatory compliance.

Liquid Ring Compressors are uniquely suited for this role due to their robustness, ability to handle wet and contaminated gases, and relatively low maintenance requirements. Their compatibility with variable gas streams and tolerance for challenging operating conditions make them particularly advantageous in oil and gas plants environments, where flaring frequency is unpredictable.

As global pressures mount to eliminate routine flaring and improve sustainability, FGRS with LRCs offer a technically viable and environmentally responsible pathway forward. Their integration into both

greenfield and brownfield oil and gas projects should be considered not only as a compliance measure but as a contribution to long-term operational efficiency and environmental stewardship.

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